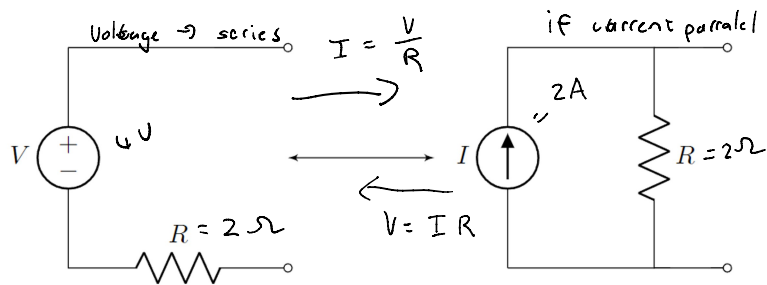


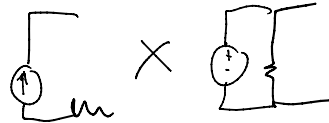
L14 - Source Transformation

Thursday, March 14, 2024 11:09 AM

Core Principles:

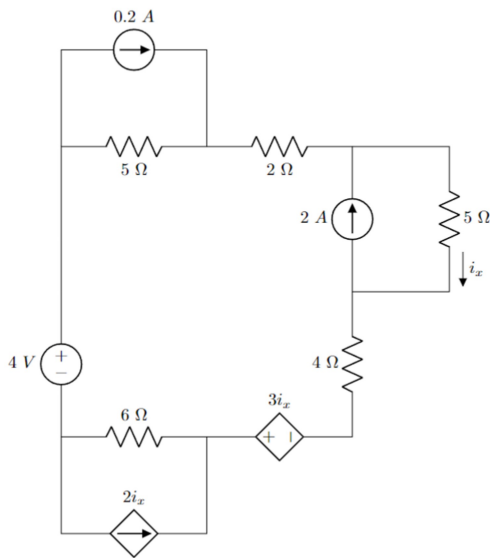


- $R_T = V_T / I_T$
- NEVER transform the measured/output/dependant values



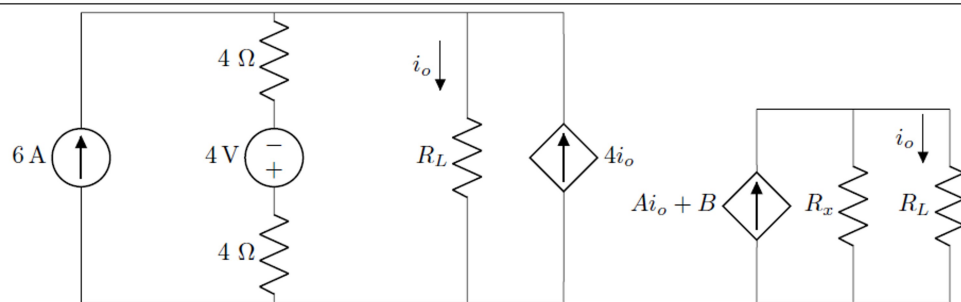
Why source transformation?

Example 1



Example 2

Simplify the circuit on the left by means of source transformations so that it looks like the circuit on the right. Determine R_x , A and B .

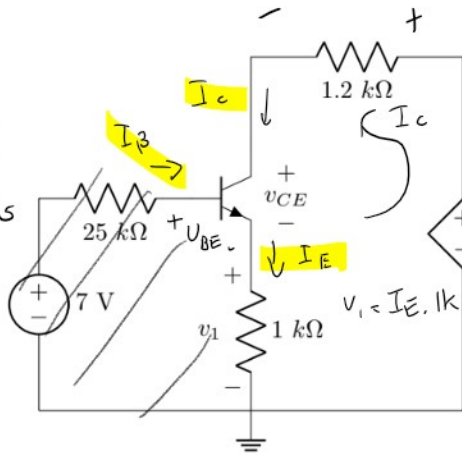


Complete while getting seated:

If $i_B = 50 \mu A$, $v_{BE} = 0.7 V$ and $\beta = 99$, solve for v_{CE} .

Transistor

- 1) find I_B
- 2) use $I_B : I_C$ rel
- 3) mesh analysis



$$I_B \beta = I_C$$

$$I_E = I_B + I_C$$

$$I_E = (1 + \beta) I_B = (1 + 99) I_B = 100 I_B$$

KVL:

$$-3v_1 + I_C(1.2k) + v_{CE} + 1k(I_E) = 0$$

$$-3(1k(I_E)) + I_C(1.2k) + v_{CE} + 1k(I_E) = 0$$

$$-3(1k)(100)(I_B) + (99I_B)(1.2k) + v_{CE} + 1k(100)(I_B) = 0$$

$$\downarrow \quad \downarrow$$

$$50\mu \quad 50\mu$$

$$v_{CE} + 1k(100)(I_B) = 0$$

$$\downarrow$$

$$50\mu$$

09

v_{CE} [2]

find I_B